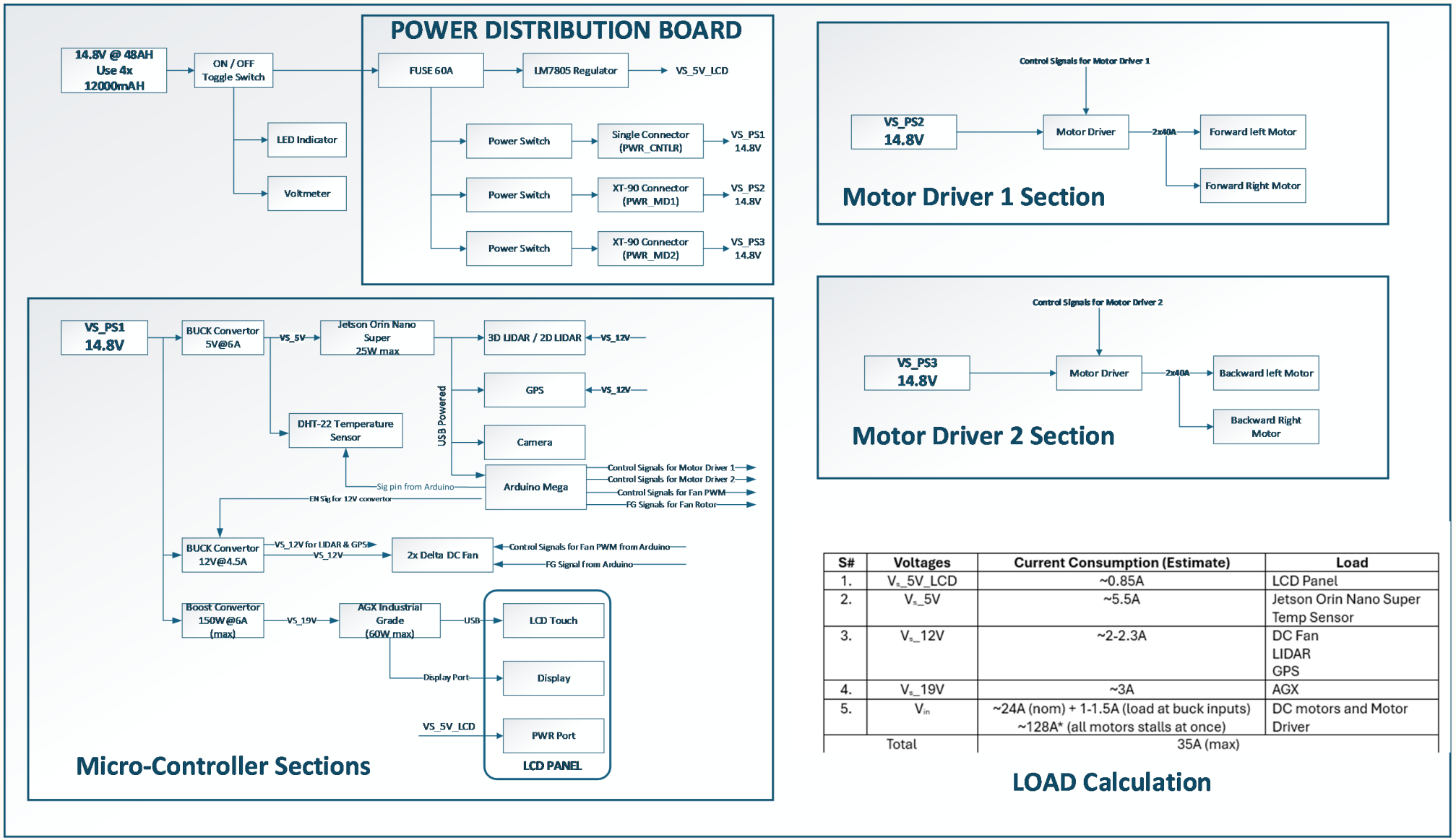
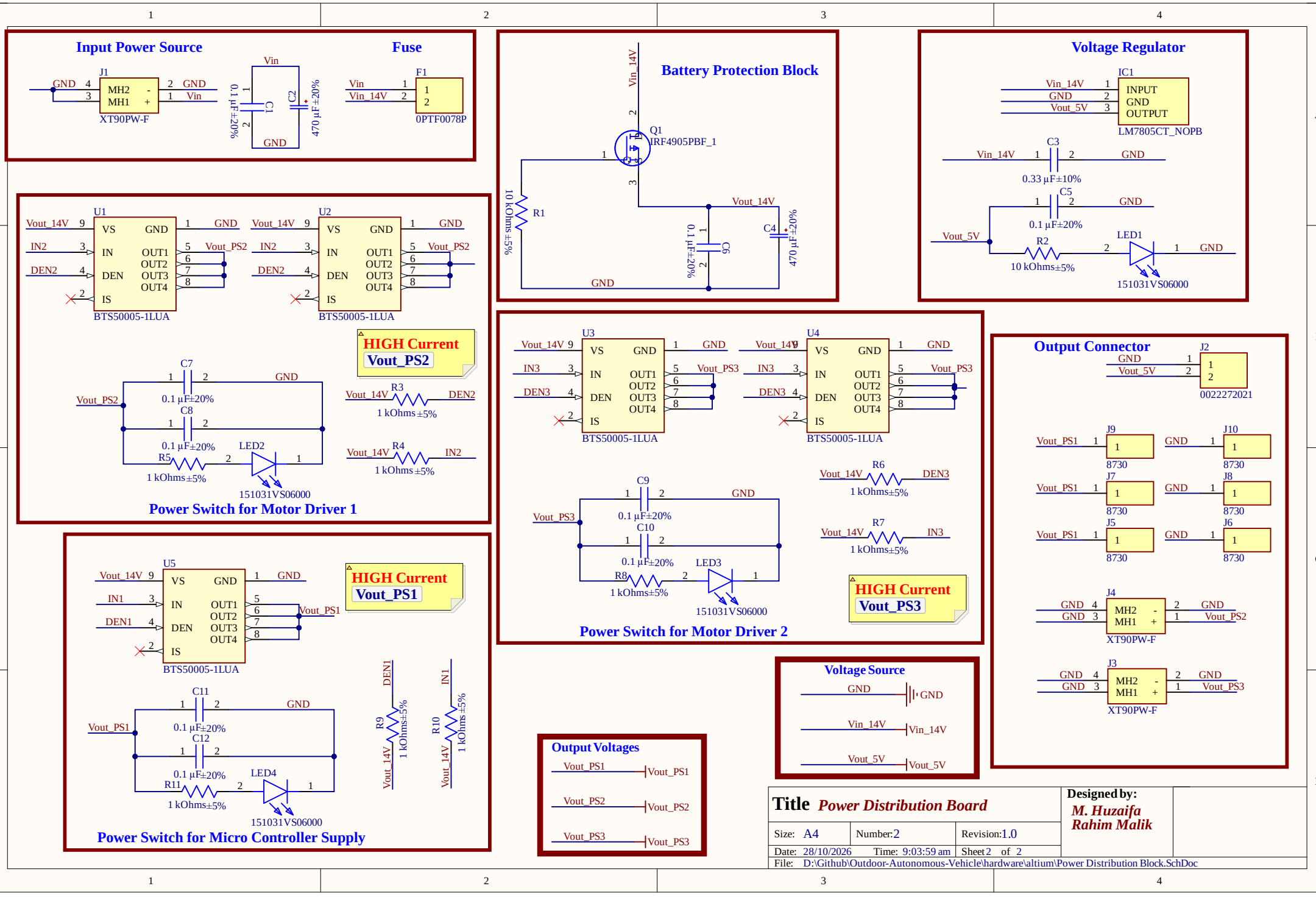
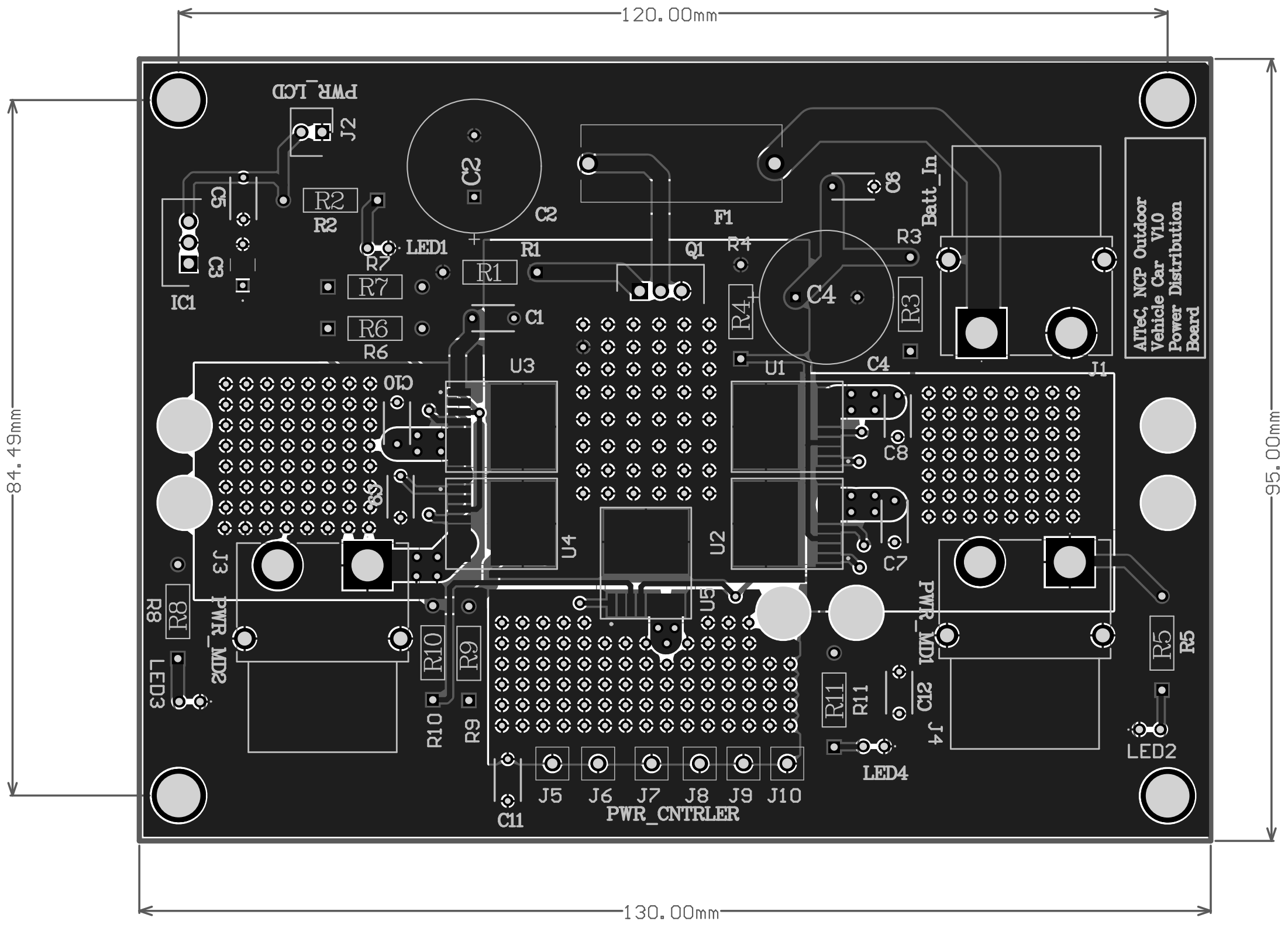


S#	Voltages	Current Consumption (Estimate)	Load
1.	V_s_5V_LCD	~0.85A	LCD Panel
2.	V_s_5V	~5.5A	Jetson Orin Nano Super Temp Sensor
3.	V_s_12V	~2-2.3A	DC Fan LIDAR GPS
4.	V_s_19V	~3A	AGX
5.	V_{in}	~24A (nom) + 1-1.5A (load at buck inputs) ~128A* (all motors stalls at once)	DC motors and Motor Driver
Total		35A (max)	

LOAD Calculation

Title *Block Diagram of System*		**Designed by:** *M. Huzaifa Rahim Malik*
Size: A4	Number: 1	Revision: 1.0
Date: 28/10/2026 Time: 9:03:59 am	Sheet 1 of 2	
File: D:\Github\Outdoor-Autonomous-Vehicle\hardware\altium\Block.SchDoc		


Input Power Source
J1
GND
MH2
MH1
XT90PW-F
Vin
C1
0.1 µF±20%
C2
470 µF±20%
GND
Fuse
F1
Vin
Vin_14V
0PTF0078P
Battery Protection Block
Vin_14V
Q1
IRF4905PBF_1
R1
10 kOhms±5%
Vout_14V
C6
0.1 µF±20%
C4
470 µF±20%
GND
Voltage Regulator
IC1
Vin_14V
GND
Vout_5V
INPUT
GND
OUTPUT
LM7805CT_NOPB
C3
0.33 µF±10%
C5
0.1 µF±20%
R2
10 kOhms±5%
LED1
151031VS06000
U1
U2
VS
GND
IN
OUT1
OUT2
OUT3
OUT4
DEN
IS
IN2
DEN2
Vout_PS2
BTS50005-1LUA
HIGH Current
Vout_PS2
C7
C8
0.1 µF±20%
R5
1 kOhms±5%
LED2
151031VS06000
R3
R4
1 kOhms±5%
Power Switch for Motor Driver 1
U3
U4
IN3
DEN3
Vout_PS3
C9
C10
R8
R6
R7
HIGH Current
Vout_PS3
Power Switch for Motor Driver 2
Output Connector
J2
Vout_PS1
J9
J10
J7
J8
J5
J6
8730
J4
Vout_PS2
XT90PW-F
J3
Vout_PS3
U5
IN1
DEN1
HIGH Current
Vout_PS1
C11
C12
R11
R9
R10
Power Switch for Micro Controller Supply
Output Voltages
Voltage Source
Title
Power Distribution Board
Designed by:
M. Huzaifa
Rahim Malik
Size: A4
Number:2
Revision:1.0
Date: 28/10/2026
Time: 9:03:59 am
Sheet 2 of 2
File: D:\Github\Outdoor-Autonomous-Vehicle\hardware\altium\Power Distribution Block.SchDoc


120.00mm
84.49mm
95.00mm
130.00mm
PWR_LCD
J2
C5
C3
IC1
R2
R2
C2
C2
F1
C6
Batt_In
R3
LED1
R7
R7
R1
R1
Q1
R4
C4
R3
R6
R6
C1
U3
U1
C4
J1
C10
C8
C9
U4
U2
C7
J3
PWR_MD2
R8
R8
LED3
R10
R9
U5
PWR_MD1
R5
R5
R10
R9
R11
R11
C12
J4
LED2
LED4
J5
J6
J7
J8
J9
J10
C11
PWR_CNTRLER
AITeC, NCP Outdoor
Vehicle Car V1.0
Power Distribution
Board

Description	Designator	Manufacturer Name	Manufacturer Part Number	Supplier	Supplier Part Number	Value	Tolerance	Rating	Quantity
0.1 µF ±20% 50V Ceramic Capacitor Z5U Radial	C1, C5, C6, C7, C8, C9, C10, C11, C12	KEMET	C322C104M5U5TA	Digikey	399-4331-ND	0.1 µF	±20%	50V	9
CAP ALUM 470UF 20% 50V RADIAL TH	C2, C4	Nichicon	UHE1H471MHD3	Digikey	493-1615-ND	470 µF	±20%	50 V	2
CAP CER 0.33UF 50V X7R RADIAL	C3	TDK Corporation	FA24X7R1H334KNU06	Digikey	445-180638-1-ND	0.33 µF	±10%	50V	1
FUSE BLOCK CARTRIDGE PCB	F1	LITTELFUSE	0PTF0078P	Digikey	F6094-ND				1
IC REG LINEAR 5V 1.5A TO220-3	IC1	Texas Instruments	LM7805CT/NOPB	Digikey	296-47192-ND	5-35V		1.5A	1
Connector: DC supply; socket; male; XT90; 40A; 500V; PIN: 2; straight	J1	Amass	XT90PW-M	TEM Electronics Component	XT90PW-M	Male			1
CONN HEADER VERT 2POS 2.54MM	J2	Molex	0022272021	Digikey	WM4111-ND				1
Connector: DC supply; socket; female; XT90; 40A; 500V; PIN: 2; THT	J3, J4	Amass	XT90PW-F	TEM Electronics Component	XT90PW-F	Female			2
TERM BINDING POST M2.5 1 PIN PCB	J5, J6, J7, J8, J9, J10	Keystone Electronics	8730	36-8730-ND	Digikey				6
LED GREEN DIFFUSED 3MM ROUND T/H	LED1, LED2, LED3, LED4	Wurth Elektronik	151031VS06000	Digikey	732-5008-ND	2.2V		20mA	4
MOSFET P-CH 55V 74A TO220AB	Q1	Infineon	IRF4905PBF	Digikey	IRF4905PBF-ND			74A	1
RES 10K OHM 5% 1/4W AXIAL	R1, R2	Stackpole Electronics Inc	CF14JT10K0	Digikey	CF14JT10K0TR-ND	10 kOhms	±5%	0.25W,1/4 W	2
RES 1K OHM 5% 1/4W AXIAL	R3, R4, R5, R6, R7, R8, R9, R10, R11	Stackpole Electronics Inc	CF14JT1K00	Digikey	CF14JT1K00TR-ND	1 kOhms	±5%	0.25W, 1/4W	9
Smart High-Side Power Switch	U1, U2, U3, U4, U5	Infineon Technologies	BTS500051LUAAUMA1	Digikey	448-BTS500051LUAAUMA1TR-ND	3.1V ~ 28V		65A	5